

The integration of digital games into teaching and learning—A unique constructivist framework

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Abstract

In today's era, the ability to create and utilize digital games in the classroom is of great importance, for both teachers and students. Yet traditional teaching methods often lack the implementation of games in general, and of digital games in particular. This study presents a unique constructivist framework for creating and integrating digital-game generators in the classroom, following four consecutive year-long courses for preservice teachers at a college in Israel. Using design-based research, year-to-year improvements (iterations) to the course were introduced, achieved through analysis of in-class discussions, semi-structured interviews, participants' reflective diaries and the generated games. The participants were required to apply the constructivist framework to a study unit in their field of expertise, during their practical school-teaching experience. The findings in this study are based on the participants' teaching experience in secondary schools in three core subjects: English as a second language, mathematics and history. The developed consecutive framework comprises three core elements: the game, the pedagogical and didactic core, and the knowledge—all of which are important when integrating digital-game generators into the classroom. The course and framework addressed in this study could be applied in additional teacher-training programs, to increase digital-game generation and related skills in classrooms.

KEYWORDS

constructivist framework, digital games in classrooms, digital-game generator, preservice teachers

Practitioner notes

What is already known about this topic

- Creating and implementing digital games in the classroom has gained momentum in recent years.
- Early exposure of preservice teachers to digital games during the teaching training phase may increase their use in the educational field.
- The developing and integration of digital games in the classroom, based on the constructivist approach, allows students to create games using existing game generators, while expressing their insights and knowledge in a personal and creative manner.

What this paper adds

- Introduces an innovative framework integrating educational game generation into teacher training, addressing key pedagogical and content knowledge components.
- Demonstrates the successful application of this framework across multiple disciplines (English, mathematics and history) in secondary education.
- Presents a longitudinal study showcasing the iterative development and refinement of a game-based learning approach in preservice teacher education.
- Provides evidence-based insights into the effectiveness of incorporating digital game creation as a constructivist learning tool in teacher preparation programs.

Implications for practice and/or policy

- The constructivist framework helps expose preservice teachers—and in early stages of their teacher training—to experimenting with different types of digital games and different methods for integrating them into their teaching.
- Such integration of digital games into their teaching may promote meaningful learning, with an emphasis on the development of important 21st-century skills.
- In light of the proposed model, the study recommends promoting the topic of digital games in teacher training, for the benefit of the preservice teachers' future students.

INTRODUCTION

Researchers emphasize the contribution of educational games in teaching and learning (eg, Boyle et al., 2016; Connolly et al., 2012; Papadakis, 2018; Yeh et al., 2019), with the professional literature suggesting that these should be increasingly integrated into the curricula (Gee, 2003; Hussein et al., 2019; Prensky, 2008). Implementing educational games in classrooms allows learners to develop important 21st century skills, including problem-solving, research capabilities, self-regulated learning and creativity—which are greatly lacking in traditional learning (Groff et al., 2010; Kahila et al., 2020; Klopfer et al., 2009). Such skills are necessary for becoming an integral part of society in general and of the employment market in particular—in today's era (Collins & Halverson, 2018). Yet to do so, perceptions of games in the field of education must be improved, to enable a transition from traditional learning methods to more adapted and relevant learning methodologies such as games.

When addressing games in education, extensive research depicts the positive effects of digital games; yet these too are still relatively uncommon in classrooms (Papadakis, 2018; Sanina et al., 2020; Vanbecelaere et al., 2020). For example, studies indicate that although

teachers are aware of the educational benefits of digital games (Hayak & Avidov-Ungar, 2020), they rarely incorporate them into their teaching (Avidov-Ungar & Hayak (2021); Fokides & Kaimara, 2020; Stieler-Hunt & Jones, 2019). To alter this, teachers may need to possess a better understanding of the functionality and advantages of digital games, in educational contexts in general, and in relation to their specific curricula in particular (Rüth et al., 2022). Moreover, teachers who do integrate digital games into their teaching tend to focus on the students' lower order thinking strategies—rather than higher-order thinking strategies—for example, by enabling the learners to practice the newly acquired knowledge through closed-ended or multiple-choice questions (Eyal et al., (2023). Hayes and Ohnberger (2013) point out that the exposure of teachers to digital gaming—as early as during the teacher-training phase—could lead to the increased use of digital games in the classroom.

This study presents a constructivist framework for incorporating the generation of digital educational games in teacher-training programs, as a means for enhancing a range of 21st century skills in their future students.

Utilizing digital-game generators in teaching and learning

While creating games in the past required great time and effort, a variety of digital-game generators (ie, platforms and applications) are readily available today, enabling users to create games via a generic base template, upon which different content can be uploaded (Baytak & Land, 2010). Using digital-game generators, educators can develop a range of tailored, engaging and interactive game-based experiences—directly aligned with their curricula and goals, thereby seamlessly enriching the K–12 learning continuum, that is from kindergarten to the end of high school (Magen-Nagar et al., 2019; Prensky, 2008). Specifically, researchers emphasize the contribution of digital games to the students' creativity and design thinking (eg, Domínguez et al., 2013; Ejsing-Duun & Karoff, 2015; Hwang & Liao, 2014; Prensky, 2008). By creating such games, students move from being in the position of responder, practitioner and trainer, to that of developer and creator, that is they transition from consumers of knowledge to producers of knowledge (Giannakos & Jaccheri, 2018).

Two approaches have emerged regarding teaching and learning through digital games: the instructional approach and the constructivist approach (Kafai, 2006). With the former approach, the students *play* a game that has been prepared by a teacher or other expert, as a means for practicing specific content. In the latter, the students *create* a game themselves, using a digital generator, thereby expressing their own creativity, insights, and knowledge regarding the given topic. The constructivist approach is rooted in the constructivist theory of learning, which emphasizes *learning by doing*, that is enhancing the students' personal involvement in the learning process. This approach also fosters creativity, problem-solving and deeper engagement with the subject matter (Griffin & Richard, 2023). The game environment offers sets of rules and intuitive tools that facilitate the learners' understanding of the environment in a relatively fast manner, while enabling them to build game activities that are tailored to their specific needs and target audience. Finally, unlike learning through games that are created by experts, and that are limited to specific skills or areas of knowledge, learning through game *creation* can be beneficial for any subject and at any level (Kafai & Burke, 2015). As such, the constructivist approach was applied in the current study.

The constructivist approach to digital-game generation

According to Kafai and Burke (2015), the constructivist approach may not be widely used in elementary schools, since educators regard it as requiring great technical effort—especially

in the context of creating digital games that require programming. Yet today, a wide variety of game generators exist, providing readily accessible platforms for creating quizzes, for example, without the need for any programming skills—such as Triventy, Quizlet and Kahoot! (Plump & LaRosa, 2017; Wang & Tahir, 2020; Yeh et al., 2019). Kahoot!, for example, is a fun, engaging and motivating digital game, that has been found to positively affect learning outcomes and classroom dynamics, while decreasing student anxiety (Wang & Tahir, 2020). An additional benefit can be seen in the process of developing the specific content for the game, which constitutes a framework for higher-order thinking skills and requires self-regulated learning among students. Yet this approach is not without challenges, including technical difficulties, the need to ensure that all questions and answers are correctly formulated, and time-based scoring that may be less suitable for some students (Wang & Tahir, 2020).

Yet generating digital games enables learners to connect more with one another (Kafai & Burke, 2015), as educators are able to create a more collaborative learning environment where learners share the same interests (Hughes-Roberts et al., 2023; Ratnasari et al., 2023). Moreover, digital-game generating has been shown to improve learners' digital competencies in other school activities and allows learners to take different perspectives during the design process (Dishon & Kafai, 2020; Hughes-Roberts et al., 2023). While the constructivist approach focuses on the students' active and self-regulated role, it also emphasizes the teachers' role. Indeed, teachers must observe their students' work, identify their difficulties, and recognize the underlying reasons for such issues—while providing them with the necessary scaffolding for overcoming such obstacles (Jong, 2016). Instead of simply memorizing and applying the acquired material in the traditional manner, students should be encouraged to apply their accumulated knowledge through the creation of games—as a means for improving their learning performance (Kiron et al., 2020; Stieler-Hunt & Jones, 2019; Yeh et al., 2019).

When teachers and students create and implement digital games in the classroom, this serves as an active means of learning, comprised of both processes and products and incorporates constructivist principles—such as transferring the learning responsibility from the teacher to the student (Weitze, 2015; Yang & Chang, 2013; Yiannoutsou et al., 2014). Active learning develops the joy of creation, as well as original and critical thinking (Duun-Ejsing & Karoff, 2015; Prensky, 2008; Yang & Chang, 2013). Furthermore, learning through the joint creation of digital games empowers students and increases their involvement in the learning process (Beça et al., 2020; Hwang & Liao, 2014; Li, 2018; Weitze, 2015; Yang & Chang, 2013). In their study, Li (2018), for example, found that teachers who designed games tended to adopt more compelling storylines with clear educational connections—based on the constructivist theory. Yet the successful generation and integration of digital games in the classroom—by students or teachers—is greatly dependent on the teachers' related theoretical and practical knowhow (Ishak et al., 2023). We therefore claim that this topic should be introduced as part of the teacher training process.

Frameworks for creating and integrating digital games in the classroom

According to Koehler and Mishra (2009), the technology, pedagogy and content knowledge (TPACK) framework suggests that the effective use of technology in the classroom is dependent on teachers understanding the relationship between the following three types of knowledge: (1) *pedagogical knowledge*, which refers to teaching approaches and methods, including the understanding of the types of difficulties that are encountered by students; (2) *content knowledge*, which refers to the particular discipline that is taught in the classroom; and (3) *technological knowledge*, which refers to the development of skills and capabilities regarding the use of technological tools. The TPACK framework emerges when all three

types of knowledge exist and interact with one another, in a variety of manners, thereby intelligently utilizing technology in education.

This TRACK framework is rooted in the teacher's pedagogical knowledge (TPK) framework, devised by Shulman (1986), who argued that the teachers' pedagogical and content knowledge should not be treated as distinct knowledge; instead, at the core of pedagogical and content knowledge lies the teachers' ability to transform *subject knowledge in general* into *subject knowledge for teaching*—to facilitate students' learning by making the knowledge accessible to them (Koehler & Mishra, 2005; Shulman, 1986). Building on both frameworks, Hsu et al. (2013) proposed the TPACK-G, a framework of technological-pedagogical-content knowledge specifically related to digital games. However, teachers often lack the necessary knowledge and skills for integrating games into their teaching, and for utilizing them as a means for implementing a range of pedagogical strategies in different fields of knowledge (Eyal et al., 2023; Denham, 2019; Foster & Shah, 2015; Hsu et al., 2017).

The literature on learning through digital games addresses various components, including planning for the integration of the game, selection and creation of the game, activation of the game in the classroom and student assessments via or following the game (Bado, 2022; Foster & Shah, 2015; Kangas et al., 2017). Foster and Shah (2015) proposed a *play curricular activity reflection discussion* (PCaRD) model for integrating of digital games in the classroom, comprised of planning, elaboration, playing, and orientation and reflection. This model offers an adaptive structure for integrating games into curricula—both existing and new. Similar stages for game-based learning were also proposed by Kangas et al. (2017), including planning, orientation, gameplay tutoring, elaboration and reflection. Another example is Bado (2022), who reviewed articles that were published over a decade on the pedagogical aspects of game integration into teaching and learning, including pre-game, game and post-game stages. In the pre-game stage, pedagogical activities included instructions of the game, lectures and worksheets; in the game phase, teachers provided scaffolding, connecting the games to the curriculum, assisting students and solving problems; finally, in the post-game phase teachers organized debriefing sessions to ensure that the gameplay translated into learning outcomes for the students. Yet the articles reviewed in this paper focused more on off-the-shelf games than on the generating of novel games. In addition, certain elements were missing from the reviewed studies, including means for evaluating the students based on or in light of the digital game, methods and teaching processes, as well as how the same games can be applied for other fields of knowledge and an understanding of how to adapt the game to the given curriculum.

Following the review of the literature presented above, research is lacking on studies on the *creating* and integrating of games in the classroom, especially digital ones. Moreover, studies are lacking on the introduction of digital-game generation and utilization during the teacher-training process. The aim of this study, therefore, is to present a unique course that was developed and implemented in a teacher-training program in Israel, on the creation and integration of digital games—as a means for enhancing innovative teaching methods in the classroom, while breaking away from traditional teaching and learning structures. More specifically, the course addressed in this study strove to create an educational setting where teachers and students collaborate in the planning, creating, implementing, and evaluating of digital games as an integral part of the classroom processes and products—based on the constructivist framework.

This study is unique, as it examines the pedagogical and didactic processes entailed in the generating and integrating of digital games in the classroom in three areas of knowledge: English as a second language (ESL), mathematics and history in high schools. Moreover, the research examined the core components (didactic and pedagogical) and scaffolding that are essential for teachers, in order for them to effectively utilize digital-game generators in the classroom. Finally, the didactic framework proposed in this article could be applied to a broad range of preservice teachers, for the benefit of the education system as a whole and for their future students in particular.

METHODOLOGY

Research procedure and participants

The current study was conducted using the design-based research strategy for investigating learning environments in their natural context. Employing this approach also contributes to our theoretical understanding of the researched problem, while developing a more generalized point-of-view (Kali, 2008). In this longitudinal study, a unique teacher-training unit was developed and conducted over four consecutive years, with each iteration improving on the previous version, based on a continuous cycle of design, implementation, assessment and improvement (Barab & Squire, 2004; Bell et al., 2004; Hoadley, 2004).

The research was carried out in the elective academic course titled, *The integration of digital games into teaching and learning*. This course, that was held annually between 2018 and 2022 at a college of education in Israel, was part of a B.Ed. teacher-training program. The scope of this year-long (two-semester) course was 60 academic hours. Based on the constructivist approach, the preservice teachers who took part in the course learned about the generating and integrating of digital games in the classroom, and were then required to implement this training in their practical teaching experience in secondary schools. The courses were conducted by the authors of this study.

The preservice teachers who participated in one of the four courses were of diverse backgrounds, including age, gender, place of residence and discipline studied at the college. Such diversity enables generalization of the didactic framework proposed in this article to a broader range of students. About 28–34 students participated in each course, resulting in a total of 124 students (73% female) in the entire study (Table 1).

The study was approved by the authors' affiliated academic institution Levinsky-Wingate Academic Center, a teacher education college in Israel. In each course, the learning process took place in three consecutive phases, whereby the preservice teachers: (1) learned to use digital-game generators and created a range of educational games via these generators. This was performed in line with a range of learning theories, with an emphasis on the constructivist approach; (2) planned activities for integrating the *generating* of digital games into their practical teaching experience, while considering the elements of the pedagogical framework described above; and (3) assimilated the framework and created games together with their students. Each course was taught based on Kolb's (1984) experiential learning cycle, which includes: experiencing as learners, reading theoretical literature, investigating relevant digital applications, planning the teaching activity in the classroom, experiencing the teaching methods and conducting a meta-cognitive discussion that includes assessments and reflections (Figure 1).

In addition, an iteration was performed following each annual course, which included significant changes to its constructivist framework. In each iteration, the preservice teachers experimented with generating games in the classrooms, and conveyed challenges and insights regarding this implementation. Following each course, the authors introduced

TABLE 1 Year and number of students per course.

#	Course year	No. of students	Comments
1	2018	32	
2	2019	28	
3	2020	30	* Part of the course was conducted via Zoom due to Covid-19 restrictions
4	2021	34	
Total		124	

improvements into the framework, thereby creating a new iteration, based on the process depicted in [Figure 2](#) (Kali & Ronen-Fuhrmann, 2011). The specific challenges and solutions regarding each iteration are presented in [Table 2](#).

Tools and data sources for creating each new iteration

To enhance the course framework and create improved iterations, the researchers utilized the following four tools: (1) course discussions; (2) interviews; (3) the preservice teachers' products (ie, digital games) and (4) preservice teachers' reflective diaries. First, classroom discussions were documented and evaluated by the researchers, in relation to the preservice teachers' implementation of the framework in their classrooms. Prior to each course session, the participants were required to read certain theoretical materials and prepare notes. During

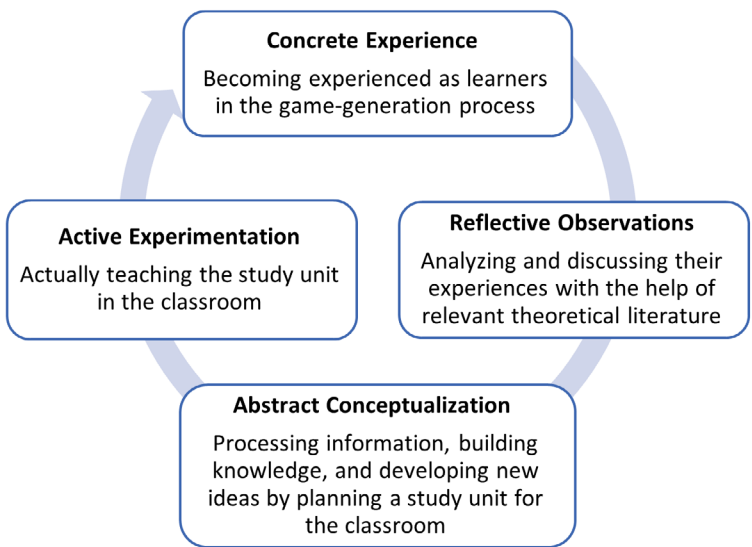


FIGURE 1 Experiential learning cycles on the course, based on Kolb (2014).

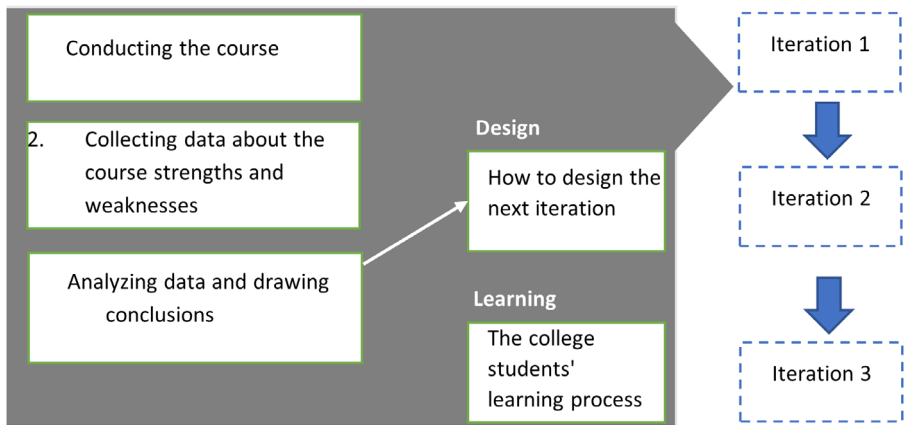


FIGURE 2 Process for creating each iteration of the *integration of digital games into teaching and learning* course.

TABLE 2 Changes to the course following each iteration.

Iteration no.	Course year	Challenges that arose from the preservice teachers	Design decisions for the following iteration
Initial course	2018	–	–
Iteration 1	2019	1. The lecturers did not provide an explanation as to why a specific game should be developed 2. Lack of group work for game planning and generation 3. Lack of choice between digital-game generators 4. No rigour in the decision-making process 5. Lack of development at the collaborative stage (ie, end-of-game development)	1. Building the development-process stages 2. Expanding on theoretical knowledge regarding cooperative learning 3. Implementation of game-application databases 4. Creating an explicit decision-making process 5. Introducing an additional stage for peer-game model
Iteration 2	2020	1. Difficulty assessing the learning process 2. Difficulty choosing age-appropriate game generators	1. Adding assessment indicators 2. Investigating game generators for older children
Iteration 3	2021	1. Incompatibility with diverse opinions 2. Students need to self-assess	1. Making the model adaptable to diverse opinions 2. Adding a self-assessment process

the class, each participant was given a set timeframe in which they could offer feedback. The preservice teachers were also required to conduct a reflective diary regarding their relevant practical teaching experience that entailed generating digital educational games. These diaries were completed in relation to planning, implementation and assessment, as well as reflective insights about the teaching and learning process that took place in the class that they taught ($n=124$). (Appendix A presents the reflection guidelines; Appendix B provides an example of a reflective-diary entry.)

Additionally, 22 in-depth semi-structured interviews were conducted with a sample of participants from various years, as a means for further understanding the preservice teachers' pedagogical experiences and challenges following their implementing of the given framework (Appendix C). The interviews were conducted via the Zoom platform, and with the participants' consent, were recorded and transcribed. Finally, the researchers examined some of the participants' products ($n=48$), that is the digital games that they and their students generated during their practical teaching experience (Appendix D).

Data analysis and processing

To ensure a robust and multifaceted analysis of the qualitative data gathered in this study, a triangulation approach was employed, integrating information from various sources. In other words, this systematic analysis process was conducted based on a multi-dimensional approach, beginning with the creation of study-unit templates for the benefit of the preservice teachers to use in their practical work experience. These templates were based on a conceptual framework which incorporated constructivist learning stages, and that were initially and deductively developed by the preservice teachers. Each template was then iteratively refined through rigorous open-coding and axial-coding processes (Strauss & Corbin, 2004), including continuous inductive comparison between items and units. For

course discussions, open coding involved transcription, initial reading, identifying themes and categorization. For course discussions, open coding involved transcription, initial reading, identifying themes and categorization. Interviews underwent axial coding, where key concepts were identified, relationships established and core categories developed. Two researchers conducted this coding and analysis independently, with disagreements resolved through discussion. Preservice teachers' products were analysed using cross-case comparative analysis (Borman et al., 2012), involving individual and cross-case pattern identification, comparison, and synthesis. Researchers guided participants in comparative discussions, then processed their conclusions. Reflective diaries were analysed using open and axial coding to ensure themes reflected diary patterns. This involved reading, theme identification, relationship establishment and integration with interview data.

By meticulously analysing and cross-referencing the data from these diverse sources, we were able to develop a grounded theory (Ralph et al., 2014; Schofieldy, 2000). Not only did this approach ensure the validity of the findings, yet it also enriched the theoretical construct with practical field-based insights.

FINDINGS

The preservice teachers who participated in the course specialized in a range of secondary-school subjects; yet the examples and explanations presented in this chapter are related to three core subjects: ESL, mathematics and history, as these are likely taught in most countries and as such are especially relevant. Each of the following examples depicts the activities that were defined and executed by the preservice teachers at various stages of the specific study unit, such as presenting the topic to the class, introducing a range of game generators, assigning tasks to the class—including digital-game generating—and assessing the students' work (Figure 3).

Example 1: A study unit in ESL on Present Simple, for grade 7. The study unit included five lessons. Table 3 depicts the seven stages that the preservice teacher developed for applying the unit in the classroom.

Example 2: A summary in mathematics on the properties of squares, for grade 10, after learning about quadrilateral-family properties. The study unit included five lessons. Table 4 depicts the eight stages that the preservice teacher developed for applying the study unit in the classroom.

Example 3: A study unit in history on the industrial revolution for grade 9. The study unit included five lessons. Table 5 depicts the six stages that the participant developing for applying the study unit in the classroom.

These three examples were derived from the qualitative-data analysis that was conducted in this study, enabling the formulation of a constructivist framework for creating and integrating digital-game generators in the classroom—comprised of the three core components: game, pedagogical and didactic, and knowledge (Figure 4).

This constructivist framework provides step-by-step guidelines for integrating game generators in the classroom. The triangulated areas between the three cores (game, pedagogical and didactic, and knowledge) convey their inter-relationships, acknowledging that educators continuously act within this complex and intertwined space.

The proposed framework offers a novel approach to guide educators in implementing student-led digital game development, addressing a gap in the literature by moving beyond teacher-centric models like TPACK and emphasizing the dynamic interplay between game design, pedagogy and subject-matter knowledge. In addition, the framework's innovative aspects include the explicit delineation of the interplay between game design, pedagogical and didactic considerations, and subject-matter knowledge, which are often treated in isolation.

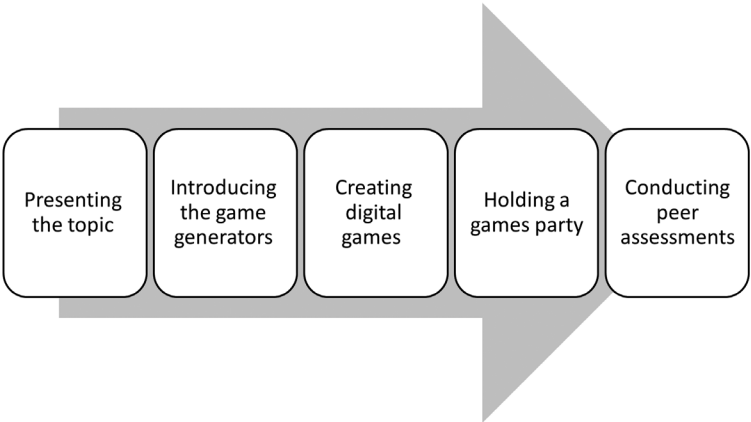


FIGURE 3 Example of main stages that were conducted by a participant.

TABLE 3 Example 1: Stages of an ESL study unit, including game generation.

Stage no.	Didactic planning
Stage 1: Pedagogical content	Presenting the <i>present simple tense</i> , including explanations, examples and a short practice task
Stage 2: Assignment allocation	Dividing the class into five groups and assigning a task to each group: Group A: Study the topic and then generate a digital game related to <i>positive sentences</i> in the present simple Group B: Study the topic and then generate a digital game related to <i>negative sentences</i> in the present simple Group C: Study the topic and then generate a game related to <i>yes/no questions</i> in the present simple Group D: Study the topic and then generate a game related to <i>WH-questions</i> in the present simple Group E: Study the topic and then generate a game related to <i>spelling rules</i> in the present simple
Stage 3: Presentation of game generators	Introducing the game generators in the class and explaining how to use each one, including: Wordwall, Quizlet, Wizerme and JEOPARDYLABS
Stage 4: Informed decision	Choosing a game generator: Each group selected a game generator and then explained why they had chosen that specific one for their given assignment
Stage 5: Game generation	Creating the game: Each group used their chosen generator to create a game, in line with their given assignment. The group members built the game at different levels, with the help of the generator functions. They then experimented with their newly generated game, ensuring that it worked correctly. The teacher and students also checked that the questions they had defined were suitable for the topic, that the answers were correct and that the game was challenging and varied
Stage 6: Game playing	Playing the generated games: Each lesson, two groups led the learning, guiding the rest of the class through the games they had developed
Stage 7: Game assessment and feedback	Assessing the game: The entire class assessed each game based on pre-defined indicators, peer assessments and in-depth class discussions

It also highlights the crucial role of supportive strategies in reinforcing these core components, fostering a dynamic and collaborative learning environment where students actively construct knowledge through game creation and play.

TABLE 4 Example 2: Stages of a math study unit, including game generation.

Stage no.	Didactic planning
Stage 1: Pedagogical content presentation	Presenting a general explanation of the quadrilateral-family properties and creating assessment indicators for the task at hand. Indicators included criteria such as question difficulty or degree of suitability between the questions and the chosen game generator
Stage 2: Assignment allocation	Dividing the class into pairs
Stage 3: Content choice	Choosing the topic: Each pair chose one type of shape
Stage 4: Game-generator choice	Choosing a game generator: Each pair chose one game generator from a number of given options and learned how to use it
Stage 5: Game generation	Each pair created a game using the selected generator
Stage 6: Game assessment	The preservice teacher and the students examined the developed games, applying a learning-from-mistakes approach. In addition, assessment indicators were created together with the students, peer assessments were conducted and in-depth discussions about each game were held in the classroom
Stage 7: Game fair	In the following lessons, the different generated games were set out in the classroom, allowing the students to experience the different games
Stage 8: Feedback and reflection	At the end of each game, the students completed a written feedback form for their peers, based on the indicators that were defined in Stage 1

TABLE 5 Example 3: Stages of a history study unit, including game generation.

Stage no.	Didactic planning
Stage 1: Pedagogical content	Introducing the topic of the industrial revolution, including relevant background material and key terminology (eg, capitalism, migration and urbanization processes). The material was divided into separate units, based on the number of teams that were to be created in the following stage. The students were asked to read texts and identify the key concepts
Stage 2: Division into groups	Dividing the class into heterogeneous groups of 4–6 students
Stage 3: Content choice	Choosing the topic: Each group chose a different text to focus on for their game generating
Stage 4: Game-generator choice	Each group chose one game generator out of four options that were suitable for trivia questions of different levels
Stage 5: The game party	Two alternatives were applied: (1) members of a given 'expert' group sat with the other groups, who were playing the former's generated game (jigsaw style); and (2) some games were played in a classroom format, where the expert group conducted their game for the entire class. The generated games were played over a number of lessons. After each trivia question, the 'expert' representative explained and elaborated on the answer to the members of the playing group or to the entire class
Stage 6: Game assessment	The different groups provided peer-feedback regarding the various games; each generating group also conducted group and individual reflections regarding the learning process

This framework could help educators implement and reflect on their use of digital games in the classroom, guiding them to ensure pedagogical and didactic best practices. Supportive pedagogic and didactic strategies, such as collaborative learning, think-pair-share and group discussions, allow teachers and students to share ideas, challenge each other's thinking, and refine their understanding—all in a collaborative manner (Lai, 2021).

Emphases, challenges and considerations for preservice teachers

During the course, a range of pedagogical issues were raised by the participants, as integrating students' digital-game generation in class requires much planning and responsibility on the teachers' part: It is the teacher who is responsible for making decisions regarding the classroom setting, including how to divide the classroom into groups, the activity timeframe, the learning materials and the digital tools. Moreover, as the students work to generate and implement the games, it is the teacher's role to guide the students throughout this process. With each iteration of the course, more emphasis was placed on these issues (Table 6).

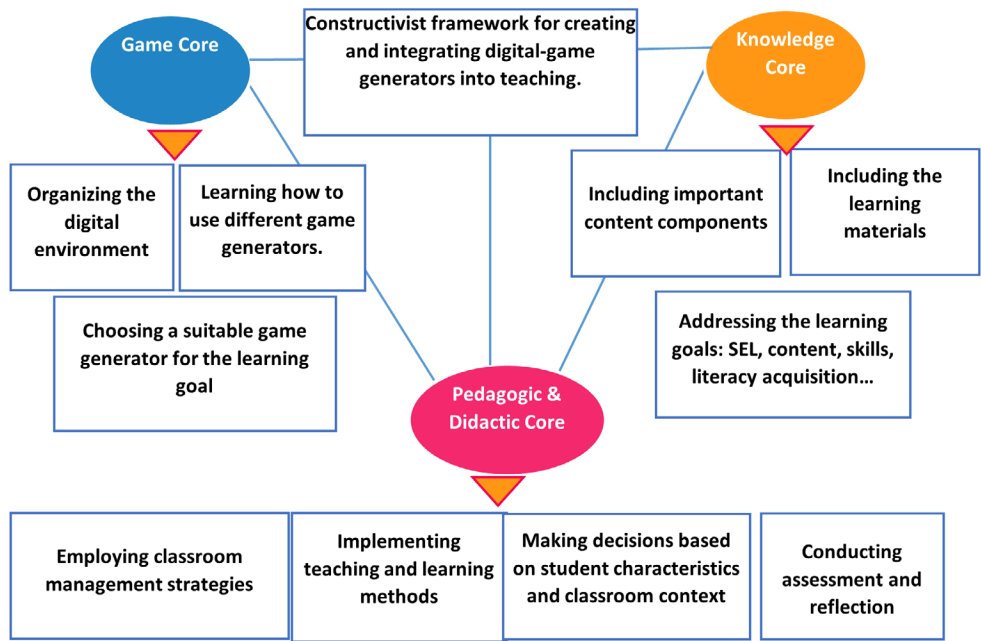


FIGURE 4 Proposed constructivist framework for generating and integrating digital games in the classroom.

TABLE 6 Pedagogical aspects when implementing digital-game generators in the classroom.

Pedagogical aspect	Elaboration
Pedagogical approach	Determining the pedagogical approach and the learning process, from start to completion of the study unit
Teacher's role	Developing, guiding, instructing and offering advice
Dividing the class into groups	Deciding on the number of groups and their composition (eg, heterogeneous or homogenous)
Learning materials	Creating and presenting information, worksheets, books, presentations, and even artificial intelligence tools (eg, ChatGPT)
Duration	The number of lessons in total and the stages for each lesson
Organizing the digital and learning spaces	Required technology, links to games, organizing the class layout in line with the lesson and classroom management
Question content	Validating the accuracy of the questions or the game components. Providing an opportunity to learn from mistakes
Assessments	Developing assessment indicators together with the students, conducting peer assessments and holding in-depth discussions about each game

DISCUSSION

The aim of this study was to present an applicable constructivist framework for implementing digital-game generation into study units in the classroom—based on a novel year-long course for preservice teachers that was developed and implemented at a college of education. The course was greatly improved over time, based on findings from the field, resulting in the initial course and three following iterations. The literature indicates the contribution of integrating digital games into teaching and learning (eg, Boyle et al., 2016; Connolly et al., 2012; Papadakis, 2018; Plump & LaRosa, 2017; Wang & Tahir, 2020; Whitton, 2014). Moreover, creating digital games based on a constructivist approach calls for a meaningful learning process, where learners make decisions and apply digital literacy skills, as well as creative thinking capabilities—all in a collaborative manner (Kiron et al., 2020; Stieler-Hunt & Jones, 2019). Yet even when teachers utilize digital-game generators in the classroom, they tend to prioritize students' enjoyment over higher-order thinking skills (Foster & Shah, 2015; Hsu et al., 2017). We therefore recommend exposing preservice teachers, who are still in training, to advanced models for integrating game generators into teaching and learning—as presented in the current study.

This research builds on constructivist learning theories, by proposing an innovative framework for integrating digital game-based learning and activities across disciplines. While prior work has explored games for specific subjects (eg, Denham, 2019; Hussein et al., 2019), our model provides a comprehensive, theory-driven approach for generating and implementing educational games in any discipline—in line with core pedagogical principles. Moreover, the majority of research on digital games in the classroom is oriented toward pedagogical aspects (eg, Bado, 2022; Foster & Shah, 2015; Kangas et al., 2017)—unlike the current study that focuses on the classroom-students' constructivist generation and use of digital games. For example, Foster and Shah (2015) proposed the PCaRD model for integrating digital games in this classroom, comprised of the following stages: planning, elaboration, playing and orientation and reflection. However, their model does not refer to the three constructivist aspects—game, pedagogical and didactic, and knowledge cores—as seen in the current study.

The findings are presented through three examples from three different disciplines. The first example, related to the present simple tense in *ESL*, demonstrates how constructivist practice of a given subject can be conducted, unlike traditional practice and learning methods. Despite the existence of off-the-shelf games on this topic, self-regulated learning is of the utmost important in learners (Zimmerman, 2002). The second example shows a summary phase of a study unit in *mathematics*, where digital games were integrated on the properties of squares. The phase began with the building of assessment indicators for the task at hand, with an emphasis on cultivating higher-order thinking, digital literacy and learning among students—in a climate that allows learning from mistakes and formative peer assessments. The third and final example shows the integration of digital games as a type of trivia questions on the industrial revolution in *history* lessons. The teaching process entailed the reading of texts, mapping and differentiating between content based on its relevance. In addition, teamwork skills were enhanced through collaborative learning using the jigsaw model; as dependency on others was necessary in order to complete the game tasks, this resulted in peer learning and mentoring. Indeed, collaborative learning is an important element in digital-game generation and in the constructivist approach, that connects between learners through shared interests (Hughes-Roberts et al., 2023; Kafai & Burke, 2015; Ratnasari et al., 2023).

At the same time, when it comes to teacher training, it is important to remember that not all in-service and preservice teachers are similarly situated along the continuum for integrating technology into their classroom, which ranges from vast resistance to great integration

(Hayak & Avidov-Ungar, 2020; Fokides & Kaimara, 2020; Stieler-Hunt & Jones, 2019). Additionally, the integration of digital-game generators into teaching requires relatively high digital literacy among the teachers. As such, we recommend discussing this with the teachers, to understand where they see themselves along this continuum, and how they perceive the possibility of learning through games. Researchers and educators must also address myths, such as that digital games lead to addiction and even violence. Finally, a focus should be placed on presenting the benefits of digital-game generators in the classroom. Without addressing such beliefs and resistance, even teachers who undergo specific training for integrating digital games in the classroom may choose to overlook their newly acquired skills and knowhow, simply continuing to teach using their traditional methods.

When providing specific training on the generation and integration of digital games in the classroom during the teacher-training processes (as depicted in this study), it is imperative that pedagogical issues and challenges are addressed as they arise. For example, how to divide the class into groups, which learning materials are necessary for the games and for the learning content, how long should be spent on generating, playing, and assessing the games, how to integrate digital-game generation into the curriculum, how to organize the physical and digital learning space, how to create and conduct formative and summative assessments, and what is the teacher's role throughout the process.

With preservice teachers, it is imperative to emphasize the benefits of the constructivist framework for integrating digital games. First, it turns the digital-game generation and implementation into an integral part of the curriculum and the learning design—in addition to serving as a means for enhancing students' enjoyment. We also claim that addressing digital games as a significant part of the students' learning process—rather than simply as a reward (eg, “If you behave well during the lesson, we'll play a game”)—could better prepare teachers for their future role as designers and implementers of learning. Additionally, incorporating digital games in the classroom promotes essential skills, positive attitudes and important values among students, in addition to their acquiring important disciplinary knowledge—in turn better preparing them for a more adapted and sustainable future.

With in-service teachers, guidance and instruction on the use of the constructivist framework can help them gain expertise in digital-game generation and application in the classroom. This could be gradually achieved, with them initially conducting the planning, generation, and integration of digital games in the classroom, and later—when they are more confident and adept—transferring this responsibility and implementation to the students, with the teachers' guidance and instruction. Furthermore, this framework can also guide teachers and instructional designers in developing more engaging and interactive learning experiences for subjects such as science, math and social studies that require problem-solving and decision-making skills. By mapping out the learning objectives of a given study unit into game mechanics and narrative elements, educators can create more immersive scenarios that foster active learning, knowledge retention and student motivation. Finally, the course and framework presented in this study could be applied by teacher-training colleges, teacher-enhancement programs, schools, policymakers and other stakeholders, as a means for supporting per-service and in-service teachers, while promoting digital game-based learning and decreasing barriers to teaching with digital games.

Design principles and guidelines for educators

Based on these findings and insights, several design principles and guidelines for educators implementing digital game generation in the classroom can be extracted. First, integrate game creation as an integral part of the curriculum, aligning game design with specific learning objectives across various disciplines to ensure that games serve as tools for

meaningful learning rather than mere entertainment. Second, emphasize the development of higher-order thinking skills and digital literacy throughout the game design process. Third, implement a constructivist approach that builds upon students' prior knowledge, experiences and individual strengths. Begin with teacher-led scaffolding, gradually reducing support as students become more proficient. Encourage active learning through hands-on game creation, leading to product-based learning where students create tangible outcomes (digital games). Throughout this process, foster student collaboration and peer learning. Fourth, recognize and leverage the interplay between the three core components of the framework: the game element, the pedagogical and didactic aspects, and the knowledge domain. This principle emphasizes the importance of balancing these elements to create a cohesive learning experience. When designing and implementing digital games in the classroom, educators should consider how the game creation process can reinforce pedagogical goals, how subject knowledge can be effectively translated into game content, and how didactic approaches can enhance both engagement in the creation process and knowledge acquisition.

These design principles foster a holistic approach that allows for a seamless integration of gaming and learning, maximizing the educational impact of the digital game creation process while supporting educators in effectively leveraging game generation to enhance student engagement, promote essential skills and facilitate deeper understanding of subject matter.

Theoretical and practical implications

The theoretical implications of this study extend our understanding of constructivist learning theory in the context of digital game-based education. By integrating game generation, pedagogical considerations, and subject knowledge, this framework offers a comprehensive model for technology-enhanced learning. Practically, this research provides valuable insights for teacher education programs and curriculum designers: It suggests a structured approach to incorporating digital game creation across various disciplines, potentially enhancing student engagement and learning outcomes. The framework's emphasis on higher-order thinking skills and digital literacy aligns with contemporary educational goals, offering a pathway to prepare students for future challenges. For in-service teachers, this model presents opportunities for professional development, encouraging a gradual shift toward more interactive and student-centred teaching methods. Additionally, the study's findings could inform educational policy, particularly in areas of technology integration and 21st-century skills development in schools.

Research limitations and future research

The research, while contributing significantly to understanding the implementation of digital game-based learning, acknowledges certain limitations. The focus on specific disciplines (English as a second language, mathematics and history) calls for further case studies to establish the framework's applicability across a broader range of subjects, such as science and literature. For example, one of the challenges in a subject like biology is the difficulty in illustrating abstract concepts such as the greenhouse effect. Future research could examine the framework's potential to contribute to the explanation of such complex topics. Additionally, the model's effectiveness warrants further investigation through mixed methods research, encompassing classroom observations, student assessments and surveys. The study also underscores the need for future research to explore the model's implications

for teacher training programs, ensuring that both preservice and in-service teachers are equipped to leverage the potential of digital game generators in creating engaging and effective learning experiences.

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CONFLICT OF INTEREST STATEMENT

The authors report there are no competing interests to declare.

DATA AVAILABILITY STATEMENT

The datasets are not publicly available due to participant privacy but are available from the corresponding author upon reasonable request.

ETHICS STATEMENT

The data collected during this study is confidential and cannot be made publicly available due to privacy concerns and the promise of anonymity made to the participants. Access to the data can be granted under specific conditions to authorized researchers upon request, subject to approval by the relevant institutional review board and in compliance with the ethical standards set forth. This research was carried out in full compliance with the ethical guidelines of Levinsky-Wingate Academic Center, with approval from the institutional ethics committee (Approval Number: 2023040301). All participants were informed of the nature and purpose of the study and provided written informed consent prior to their participation. To ensure the protection of the participants, steps were taken to anonymise the data, and measures were implemented to guarantee that no participant would be disadvantaged or identified through the information collected.

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APPENDIX A

REFLECTIVE-DIARY GUIDELINES

We ask that you reflect in writing on the following questions. You may also freely express your feelings, emotions and experiences that emerged when participating on the course and performing your practical teacher-training at the school.

- Describe the challenges and obstacles that you faced during the course at the college and during your practical teaching experience at the school.
- What would you change or do differently in relation to your teaching at the school?
- How did you feel before, during and after you conducted your practical teacher training at the school? Please elaborate.

APPENDIX B

EXAMPLE FROM A PRESERVICE TEACHER'S REFLECTIVE DIARY

As I reflect on today's lesson, where I guided my students through the intricate world of digital-game production, I find myself marvelling at their journey of discovery and creation. The classroom was abuzz with excitement and curiosity as they experimented with various game generators, each offering unique features and challenges. Witnessing their enthusiastic discussions and thoughtful decision-making, as they selected the most suitable platform for their game development, was truly inspiring. It was a moment of profound learning, not just about the technical aspects of game design, but also about collaboration and critical thinking. One insightful takeaway from this experience was observing how each student's approach to problem-solving evolved, influenced by the capabilities and limitations of the game generators. This process highlighted the importance of adaptability and creative thinking in technology-driven environments. Their collective choice of a game generator, after weighing its user-friendliness against its ability to bring their creative visions to life, demonstrated a mature understanding of the balance between technical feasibility and artistic expression.

(E. January 2021)

APPENDIX C

INTERVIEW GUIDE

1. Background questions: age, gender, place of residence, discipline studied at the college.
2. Tell us about the process of choosing the digital-game generator. How did you create a game, and how did you design a lesson plan for your class in relation to your discipline?
3. Please describe how you integrated this lesson plan into your teaching in the classroom, according to the models that were taught during the course.
4. Please reflect on your practical teaching experience: What works, what would you do differently, and do you have any other didactic and pedagogical thoughts that you would like to share?

APPENDIX D

Math game created by students using Wordwall



ESL game created by students using WisersMe

